

Multiple Choice

1. **Answer:** A

Options: A) $V_{ab}(t) = -15te^{-4t} - 15e^{-4t} + 18e^{-2t}$; B) $V_{ab}(t) = 15te^{-4t} - 15e^{-4t} + 18e^{-2t}$; C) $V_{ab}(t) = 30te^{-4t} + 30e^{-4t} - 36e^{-2t}$; D) $V_{ab}(t) = 30te^{-4t} - 30e^{-4t} + 36e^{-2t}$; E) $V_{ab}(t) = 15te^{-4t} + 15e^{-4t} - 18e^{-2t}$

Note: Correct option: A - $V_{ab}(t) = -15te^{-4t} - 15e^{-4t} + 18e^{-2t}$

1. **Answer:** C

Options: A) 1200 volts/meter; B) 5 ohms; C) 1883 volts/meter; D) 940.5 volts/meter; E) 5 volts/meter

Note: Correct option: C - 1883 volts/meter

2. **Answer:** B

Options: A) $J = 0, K = 0$; B) $J = 1, K = 1$; C) $J = 0, K = 1$; D) $J = 1, K = 0$.

Note: Correct option: B - $J = 1, K = 1$.

3. **Answer:** A

Options: A) 6; B) 12; C) 4; D) 10; E) 18

Note: Correct option: A - 6

4. **Answer:** A

Options: A) 3; B) 5; C) 2.5; D) 1; E) 2

Note: Correct option: A - 3

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: $-(2z/ho^2)u_{ho} + (1/ho^3)u_z$.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'RL Circuit'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $A + B = |473| |375|$, $A - B = |-1 -9 -1| |3 -23|$ '.

Long Form Response

11. **Answer:** 34,650 Btu. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 3/4. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key